Industry Practices: Research Case Studies

Notes for the SIT-DP module: **Developing Immersive Applications**

Created by: Chek Tien TAN

slido

Please download and install the
Slido app on all computers you use

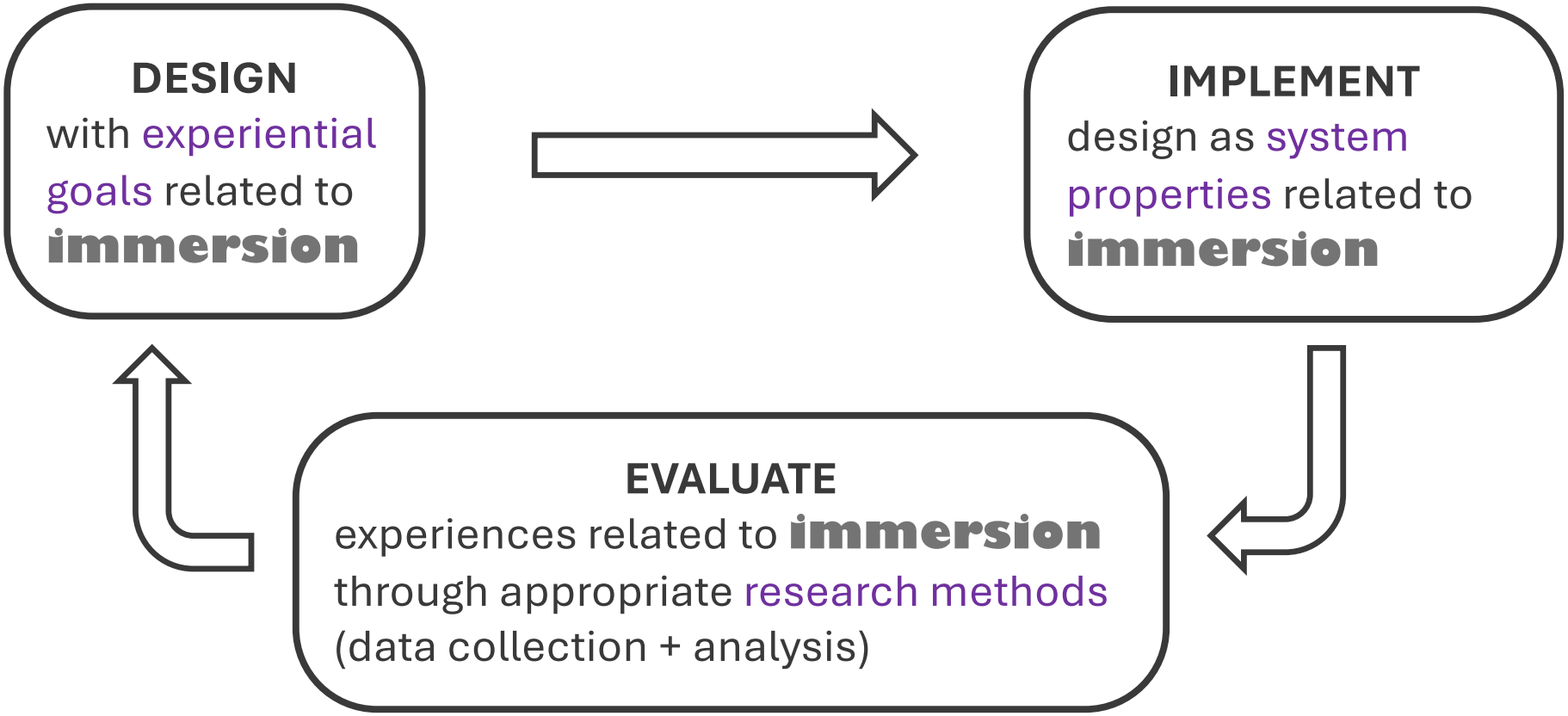
Join at slido.com
#week10-case-studies

ⓘ Start presenting to display the joining instructions on this slide.

Learning Objectives

- appreciate the design and evaluation of immersive applications through real-world case studies
- understand how design translates into implementation through the lens of immersion
- understand how user study methods are used to achieve immersive experiential goals

Developing for **Immersion**



slido

Please download and install the Slido app on all computers you use

Which of the following describes immersion from a systems perspective?

ⓘ Start presenting to display the poll results on this slide.

slido

Please download and install the Slido app on all computers you use

Which popular experiential construct(s) of immersion is/are relevant here?

ⓘ Start presenting to display the poll results on this slide.

Virtual Reality Commuting Simulator

Research Case Study 1

Designing for Immersion

It is first essential to consider all stakeholders of the application

- **Users** - the general public
  - that can come into a VR lab
  - who are interested to contribute feedback on commuting experiences
  - and perhaps curious about VR
- **Collaborator** - LTA
  - aim to have a VR system to evaluate commuting experiences
  - so as to obtain higher quality data and reduce operational costs

Designing for Immersion

- feel like actually in Singapore HDBs [PRESENCE]
- locomotion close to real walking [PRESENCE]
- clear goals and fluent performance during commute [FLOW]
- minimal nausea and disorientation symptoms [CYBERSICKNESS]

We ran workshops with the public to understand their problem and needs

Design → Implementation

feel like actually in Singapore HDBs [PRESENCE]

replicate the scale and visuals of a real HDB estate

Real-world 3D Scans

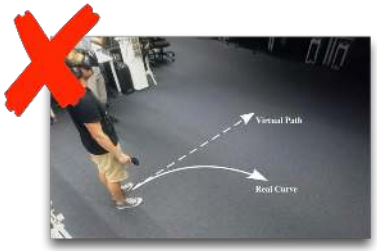
ArcGIS CityEngine

Design → Implementation

locomotion close to real walking [PRESENCE]
clear goals and fluent performance during commute [FLOW]

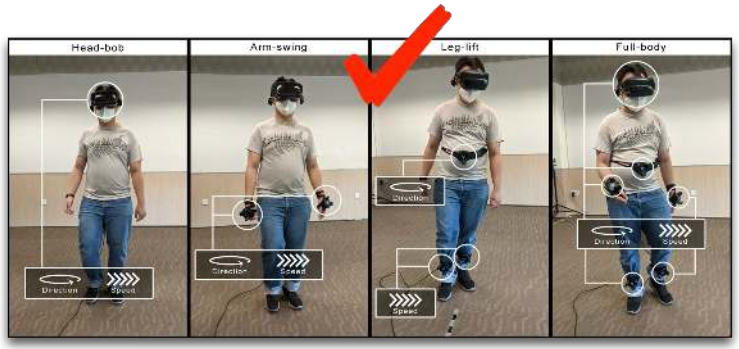
create a walking locomotion that is semi-natural yet easy to operate

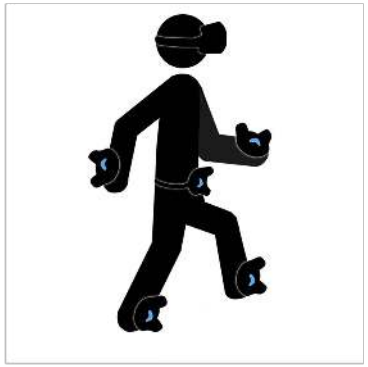
Teleportation

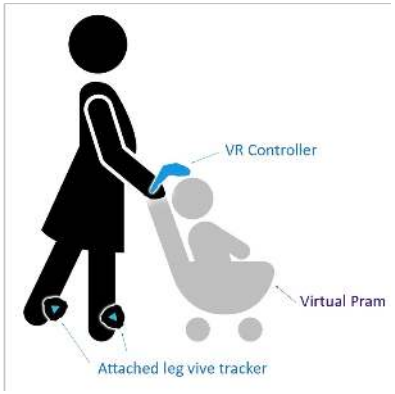


Redirected Walking in VR
(Langbehn & Steinicke, 2018)

VR "slidemills"







Design → Implementation

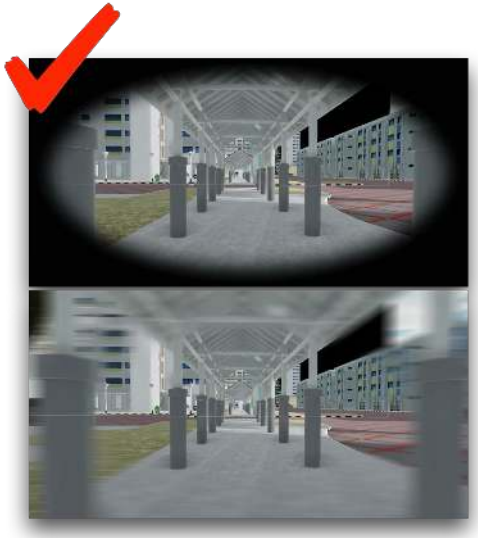
minimal nausea and disorientation symptoms [CYBERSICKNESS]

reducing common pitfalls that induce visual-vestibular conflict

Joystick

Stationary

Walking Actions



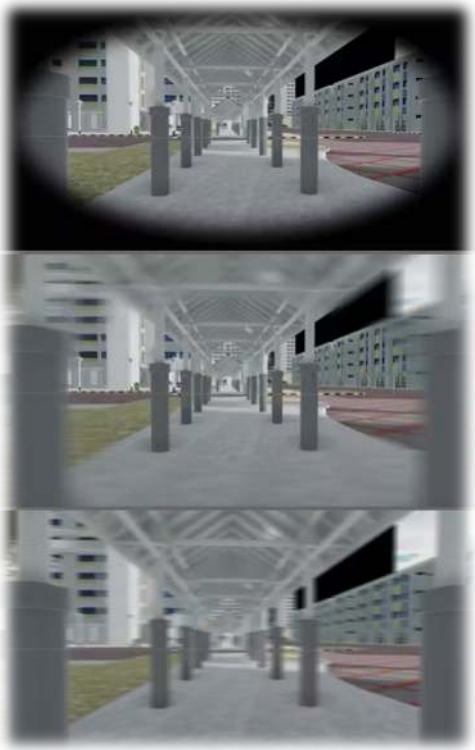
Visual mitigation
during movement

Design → Implementation

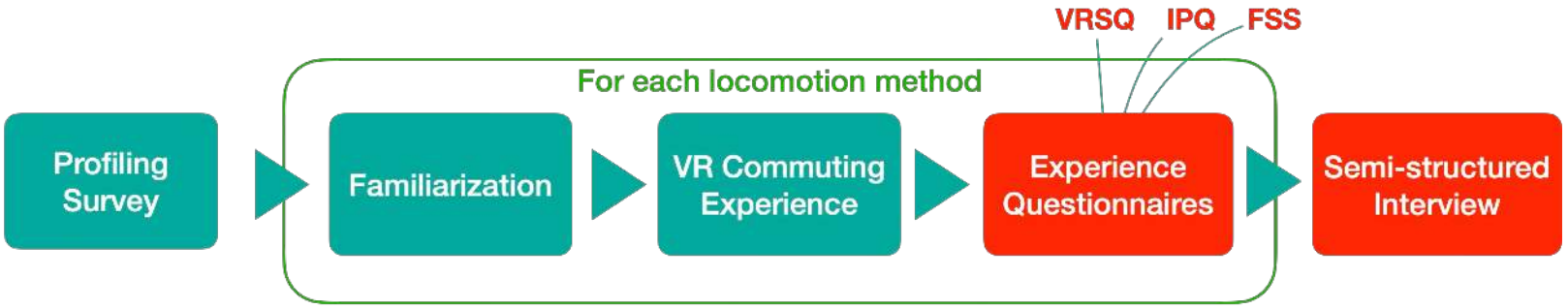
minimal nausea and disorientation symptoms [CYBERSICKNESS]

reducing common pitfalls that induce visual-vestibular conflict

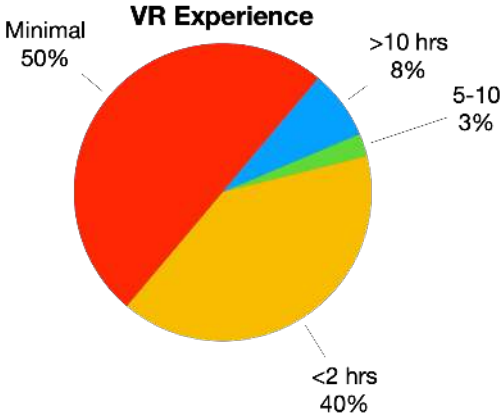
- prototyped various mitigation methods from various prior papers
  - How the Presence and Size of Static Peripheral Blur Affects Cybersickness in Virtual Reality (Lin et. al. 2020)
  - Narrative and gaming experience interact to affect presence and cybersickness in virtual reality (Weech et al. 2020)



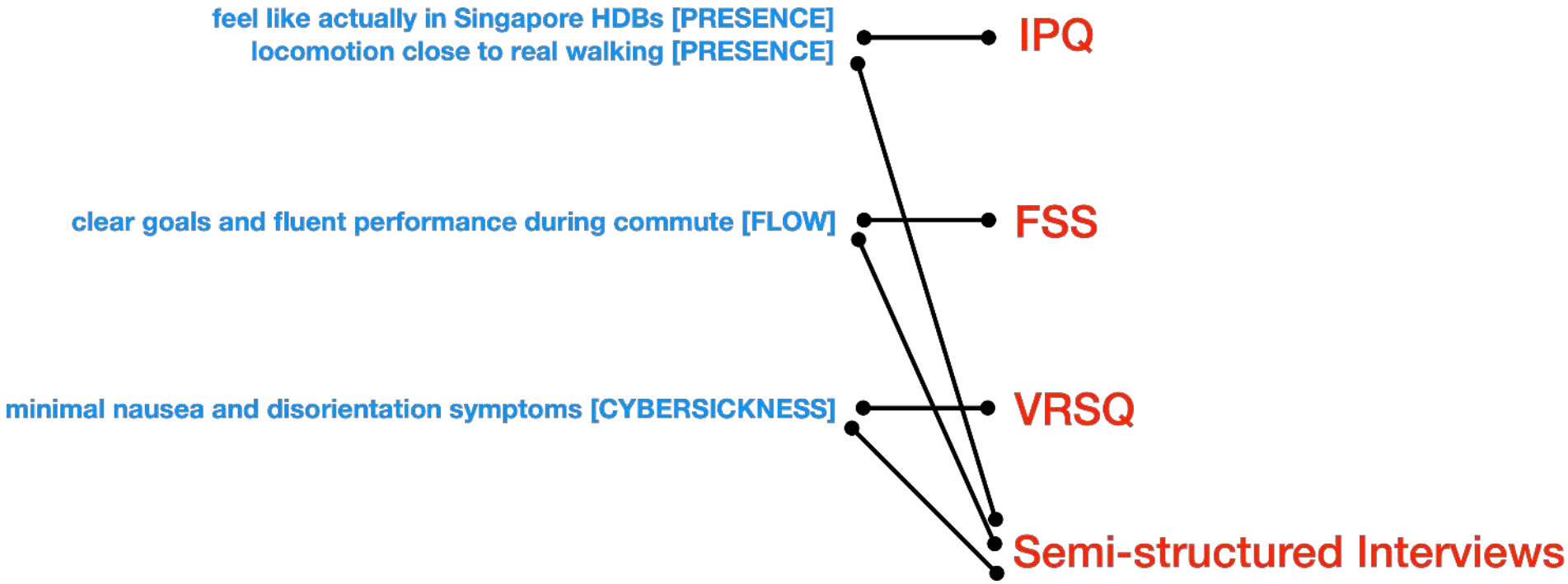
Data Collection Appropriate for Experiential Goals



40 participants **aged** 21-45

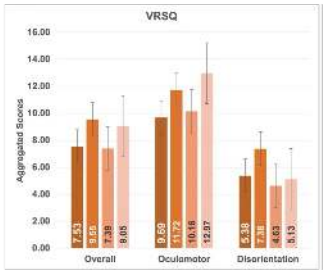


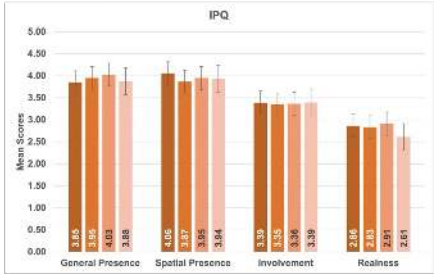
Data Collection Appropriate for Experiential Goals

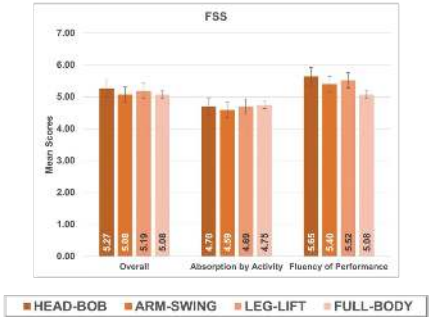


Analysing Data to Form Conclusions

		CONTEXTUAL CODES					
		HEAD-BOB	ARM-SWING	LEG-LIFT	FULL-BODY	GENERAL	Total across CONTEXTs
Minimal cybersickness with WIP exertion	FATIGUING	8	0	7	7	5	27
	EFFORTLESS	5	8	1	0	1	15
	CYBERSICKNESS	2	7	0	2	1	12
Leg-lift method facilitated most positive experiences	**POSITIVE-LOCOMOTION**	**6**	**7**	**25**	**1**	**6**	**45**
	NATURAL	4	4	18	1	0	27
	PRESENCE	2	1	6	0	5	14
	EASY-TO-FOCUS	0	2	1	0	1	4
	NEGATIVE-LOCOMOTION	**25**	**23**	**13**	**27**	**21**	**109**
	UNNATURAL	11	14	4	14	3	46
	UNSURE-HOW-TO-LOCOMOTE	6	7	8	6	8	35
	HARD-TO-FOCUS	7	1	0	2	1	11
	DISCOMFORT-EQUIPMENT	0	1	0	1	5	7
	CONSTRAINED-BY-TRACKERS	1		0	3	3	7
	CONSCIOUS-OF-APPEARANCE	0	0	1	1	1	3
Varying WIP gestures afforded by tracker positions	AWKWARD-WIP-TECHNIQUE	7	13	5	7	1	33
	ADAPTED-WIP-TECHNIQUE	7	7	5	7	2	28
	AFFORDED-BY-TRACKERS	3	8	5	2	1	19
Perception of control was important for WIP	LACK-SPEED-CONTROL	14	10	17	19	8	68
	LACK-DIRECTION-CONTROL	5	9	10	8	2	34
	LACK-MOMENTUM	1	1	2	4	5	13
Walking-"not"-in-place affects immersion	SPACE-CONSTRAINED	3	4	7	4	5	23
	REQUIRE-SPATIAL-AWARENESS	3	4	2	4	8	21
	UNSAFE	0	0	2	2	7	11
	DISEMBODIMENT	1	0	1	2	2	6
Visual qualities affect WIP motivations	UI-GUIDANCE-WORKS-WELL	0	0	0	0	27	27
	UI-GUIDANCE-CONSTRAINED-EXPLORATION	0	0	0	0	22	22
	INCLINED-TO-EXPLORE	3	2	0	1	15	21
	LACK-OF-MOTIVATION-TO-MOVE	0	0	0	0	8	8
	DESIRE-FOR-MORE-ASSETS	**0**	**0**	**0**	**0**	**45**	**45**
	DESIRE-FOR-MORE-REALISTIC-ASSETS	0	0	0	0	21	21
	DESIRE-FOR-ANIMATED-ASSETS	0	0	0	0	15	15
	DESIRE-FOR-SOUNDS	0	0	0	0	9	9







Experience Dementia in Singapore

Research Case Study 2

Designing for Immersion

It is first essential to consider all stakeholders of the application

- **Users** - the general public
  - gained knowledge to empathize with dementia
  - enhanced experience for someone with an HMD
- **Collaborator** - Dementia Singapore (dementia.org.sg)
  - aims to educate the public on dementia
  - aim for as many users as possible

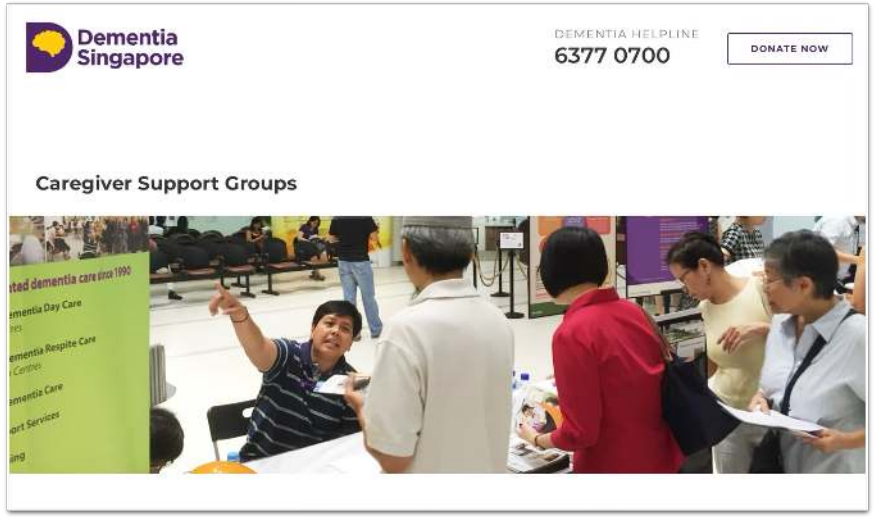
Having established the context, we craft key design goals for immersion

- improved empathy with dementia (both VR & Desktop) [EFFECTIVENESS]
- minimal physiological discomfort (both VR & Desktop) [CYBERSICKNESS]
- improved immersion in VR over Desktop [PRESENCE]

Design → Implementation

improved empathy with dementia (both VR & Desktop) [EFFECTIVENESS]

implement scenarios guided closely by caregiver domain experts



Design ➞ Implementation

minimal physiological discomfort
(both VR & Desktop)
[CYBERSICKNESS]

use minimal locomotion variability (a point-and-click experience)

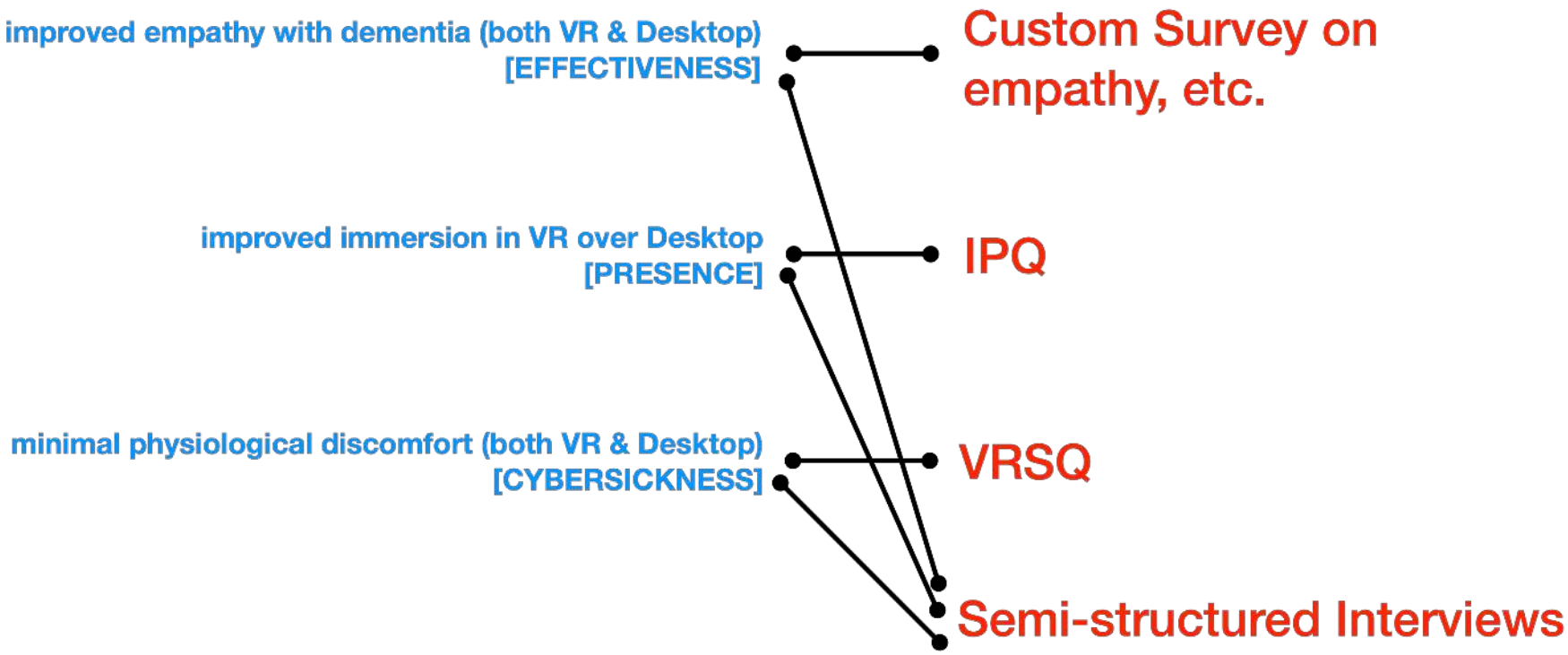
Design → Implementation

improved immersion in VR over Desktop [PRESENCE]

provide the most essential VR interaction mechanic - viewpoint control

https://dementia.org.sg/2022/03/30/experience-dementia-in-singapore/

Data Collection Appropriate for Experiential Goals



Analysing Data to Form Conclusions

Median score change in pre/post, Mann-Whitney U Test

Measure	Web-based VR		VR-headset		Z score	*p* (two tailed)	*p* (one tailed)
	Pre/post median change in score	Obs	*Pre/post median change in score*	Obs			
Dementia Attitude Scale	11	25	8	57	1.26	.21	.10
Dementia Knowledge and Attitude Scale	3	25	6	57	-3.06	.0022	.0011
Empathy and Understanding in Dementia Index	11	25	10	57	1.04	.30	.85

Mean score for Virtual reality sickness questionnaire (VRSQ), Man-Whitney U Test

Measure	Web-based VR		VR-headset		t score	*p* (two tailed)
	Mean score	Obs	Mean score	Obs		
Virtual reality sickness questionnaire (VRSQ)	3.03	25	11	27	-3.02	0.0026

Mean score for Igroup Presence Questionnaire (IPQ), and Independent t-test

Measure	Web-based VR		VR-headset		t score	*p* (two tailed)
	Mean score	Obs	*Mean score*	Obs		
Igroup Presence Questionnaire (IPQ)	-6.24	25	3.92	27	-3.86	<.001

I got to experience being a person with Dementia and learnt about some troubles I never thought persons with Dementia faced

The audio component of the web based VR made it immersive and helped invoked emotions (eg. heartbeat effect made me feel anxious while humming and the calm music made me feel calm)

I liked the VR experience because I was able to understand and empathize PWDs better in their POV

Study done by SIT Physiotherapy Students 2022

[https://www.straitstimes.com/life/what-is-it-like-to-have-dementia-experience-it-with-immersive-vr-app]

slido

Please download and install the Slido app on all computers you use

A design goal is to have minimal cybersickness in locomotion given the poor motion tracking fidelity of my phone in the cardboard.

Which of the following implementations will this design translate into?

ⓘ Start presenting to display the poll results on this slide.

slido

Please download and install the Slido app on all computers you use

A design goal is to have minimal cybersickness in locomotion given the poor motion tracking fidelity of my phone in the cardboard.

Which of the following data collection methods will help us evaluate this user experience?

ⓘ Start presenting to display the poll results on this slide.

slido

Please download and install the Slido app on all computers you use

A design goal in our VR Space Simulator is for users to feel like they are actually "there" in the space shuttle out in space (yes the zero-gravity kind).

Which of the following implementations will this design translate into?

ⓘ Start presenting to display the poll results on this slide.

slido

Please download and install the Slido app on all computers you use

A design goal in our VR Space Simulator is for users to feel like they are actually "there" in the space shuttle out in space (yes the zero-gravity kind).

Which of the following data collection methods will help us evaluate this user experience?

ⓘ Start presenting to display the poll results on this slide.

slido

Please download and install the Slido app on all computers you use

A design goal is for users in our VR museum gallery app to be fully absorbed in a smooth engaging experience in which they naturally know what to do in each step of the experience.

Which of the following implementations will this design translate into?

ⓘ Start presenting to display the poll results on this slide.

slido

Please download and install the Slido app on all computers you use

A design goal is for users in our VR museum gallery app to be fully absorbed in a smooth engaging experience in which they naturally know what to do in each step of the experience.

Which data collection methods will help us evaluate this user experience?

ⓘ Start presenting to display the poll results on this slide.

slido

Please download and install the
Slido app on all computers you use

Audience Q&A

ⓘ Start presenting to display the audience questions on this slide.